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Review

Application of CRISPR in Filamentous Fungi and Macrofungi: From Component Function to Development Potentiality

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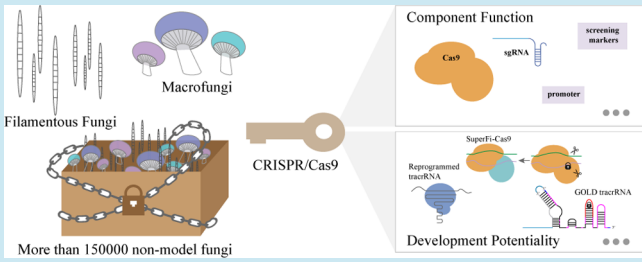
ABSTRACT: Fungi, particularly filamentous fungi and macrofungi, have a very powerful ability to produce secondary metabolites and can serve as excellent chassis cells for the production of enzymes or natural products of great value in synthetic biology. Thus, it is imperative to establish simple, reliable, and efficient techniques for their genetic modification. However, the heterokaryosis of some fungi and the dominance of nonhomologous end-joining (NHEJ) repair mechanisms *in vivo* have been greatly affecting the efficiency of fungal gene editing. In recent years, the CRISPR/Cas9 system has been applied as a widely used gene editing technology in life science research and has also played an important role in the genetic modification of filamentous and macrofungi. The various functional components (cas9, sgRNA, promoter, and screening marker) of the CRISPR/Cas9 system and its development, as well as the difficulties and potential of the CRISPR/Cas9 system in filamentous fungus and macrofungi, are the main topics of this paper.



KEYWORDS: *filamentous fungi, macrofungi, CRISPR/Cas9, homologous recombination, gene editing*

Based on morphological characteristics, fungi can be categorized into three groups: yeasts, filamentous fungi, and macrofungi (mushrooms). In general, filamentous fungi can form branched mycelia, while macrofungi generate fruiting bodies. In contrast to unicellular yeasts, filamentous fungi and macrofungi are multicellular eukaryotes, and the majority of them exhibit the characteristics of heterokaryosis, containing two or more genetically distinct nuclei within the same mycelium or spore of a fungus at a specific growth stage, which makes molecular genetic manipulation more difficult.[1−4]

Filamentous fungi are widely used in the production of numerous fermentation products because of their metabolic diversity and high efficiency in protein secretion. For example, strains belonging to *Aspergillus* and *Trichoderma* are extensively used for the production of xylanase and cellulase.[5,6] Importantly, many secondary metabolites with important biological activities produced by filamentous fungi or their derivatives have been developed into drugs for clinical use, including penicillin and statins.[7−9] The most developed macrofungi like edible mushrooms (*Lentinula edodes*, *Agaricus bisporus*, and *Flammulina filiformis*) and medicinal mushrooms (*Ganoderma lucidum* and *Cordyceps militaris*) have active ingredients such as polysaccharides, terpenoids, alkaloids, and unsaturated fatty acids that have great potential application value.

To investigate or modify the functions of various genes in filamentous fungi and macrofungi, genetic manipulation including nucleotide deletion, substitution, or insertion of target DNA sequences at the genomic level is the most commonly used strategy. Early genome-editing efforts were based on endogenous cellular homologous recombination repair pathways. In order to use this strategy for genome editing, exogenous DNA sequencing must have homology with the target genome DNA site. In addition to inefficient editing using spontaneous homologous recombination, this method could also induce unexpected genome editing events, where exogenous DNA sequences bind to the genome more frequently at random sites than at expected sites.[10] The key to overcoming this limitation is to found that the efficiency of homologous recombination at the break site can be significantly enhanced by double-strand breaks (DSBs) at the target DNA sites in the genome,[11] which is known as the "homologous directed repair" (HDR) strategy. It is worth noting that nonhomologous end connections (NHEJ) also occur at the DSBs sites, which are usually more effective than HDR, resulting to random insertion and deletion (indels) of nucleotides at the target site.[10] Although NHEJ mediated genome editing is useful for gene disruption, it is also an

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unnecessary byproduct when precise genome editing is required. Accordingly, artificial specific nuclease technologies, including zinc-finger nucleases (ZFNs),[12] transcription activator-like effector nucleases (TALENs),[13] and CRISPR/Cas9 (clustered regularly interspaced short palindromic Repeats/CRISPR-associated protein 9) system technologies, gradually emerged. These endonucleases are capable of causing double-strand breaks at targeted DNA sites in the genome, followed by gene editing via two intracellular repair pathways, NHEJ and HDR. Compared with ZFNs and TALENs genome editing technologies, the CRISPR/Cas9 system is a relatively simple construction strategy with higher fidelity and multiple target recognition capability.

In this paper, the functional components of the CRISPR/Cas9 system used in filamentous fungi and macrofungi up to now are summarized, and the difficulties and potential of the application of the CRISPR/Cas9 system in filamentous fungi and macrofungi are discussed.

1. THE WORKING MECHANISM AND DEVELOPMENT OF THE CRISPR/CAS9 SYSTEM

Originating from an adaptive immune mechanism in bacteria and archaea, the CRISPR/Cas system has evolved into the most efficient genome editing system in recent years due to its ability to specifically cleave target sequences by guide RNA.[6] Currently, among the three types of CRISPR/Cas systems that have been identified, the type II CRISPR/Cas9 system has simpler components and is the most widely used because this system relies on a single protein nuclease to achieve DNA cleavage.[14] CRISPR/Cas9 system includes Cas9 endonuclease, crRNA (CRISPR associated RNA) and tracrRNA (trans-activating crRNA).[15] Subsequently, the researchers further modified and optimized the system by linking crRNA and tracrRNA into single-guided RNA (sgRNA). Guided by sgRNA, the Cas9 protein recognizes specific DNA duplexes and performs DNA cleavage to generate DSBs (Figure 1C). The 5′ end of crRNA is a 20 nucleotides long sequence that specifically binds to the prototype spacer sequence (protospacer) of the target gene, which is responsible for identifying the target gene and requires the presence of a segment of the protospacer adjacent to the motif PAM (NGG) of the target gene.[16] The 5′ end of single-stranded tracrRNA is complementarily paired with a segment of sequence at the 3′ end of the crRNA, and the other segment forms two hairpin structures.[17] Thus, each gRNA consists of a target-specific spacer sequence and a constant part (gRNA scaffold), which consists of distinct conserved motifs, including a nexus and two hairpins (Figure 1D). In sgRNA, crRNA is used to match with the target sequence, while tracrRNA is responsible for binding and activating Cas9 protein.[18]

In recent years, with the deepening study of the CRISPR/Cas9 system, this system has become more and more mature in the application process, and the upgrading of cas9 protein and sgRNA in the CRISPR/Cas9 system has also emerged one after another. For example, since the target gene is aimed only through the 20 nt sequence, it is often the case that the Cas9 enzyme is still activated when 1 or 2 bases are mismatched, leading to a potential possibility of off-target. To address this issue, Bravo et al.[19] discovered the structural mechanism behind off-targeting in the CRISPR/Cas9 gene editing system. The Cas9 protein tightly holds the mismatch region through a finger-like structure, thus stabilizing the RNA-DNA duplex to behave like a correct pairing. Based on this groundbreaking

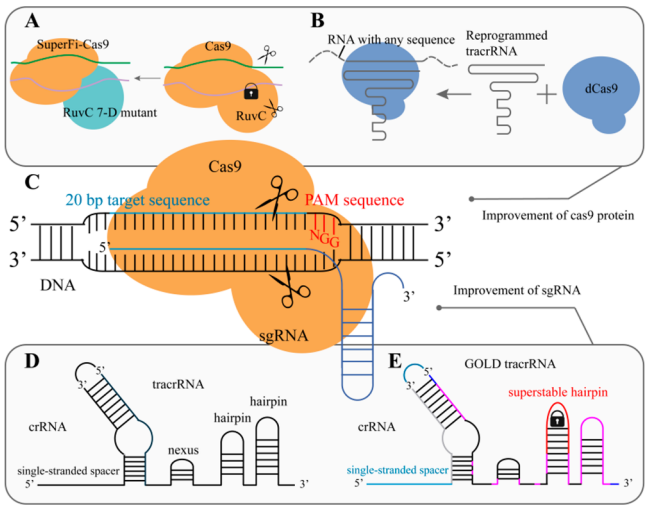

Figure 1. Schematic illustration of CRISPR-Cas9 genome editing. (A) Mutations in the RuvC loop (corresponding to SuperFi-Cas9) inhibit formation of the kinked conformation and subsequent cleavage with mismatches at the PAM-distal end. (B) Reprogramming the crRNA-tracrRNA hybridization sequence can hijack specific RNA molecules *in vivo* and *in vitro* to activate CRISPR function. (C) sgRNA-mediated Cas9 protein can bind to the target sequence of selected sites and cut DNA double strands. (D) gRNA (hybridized crRNA: tracrRNA) from *S. pyogenes* for a canonically folded gRNA. The spacer sequence, as well as the structural motif nexus, two hairpins are shown. (E) Structure and chemical modifications of the genome editing optimized locked design (GOLD) tracrRNA. Locked tracrRNA (t-lock) gRNA design carried an elongated first hairpin with a superstable loop which should serve as a gRNA folding nucleation site and prevent misfolding.

discovery and insight, they redesigned the Cas9 enzyme (SuperFi-Cas9) with a finger-like structure partially modified to stay away from DNA (Figure 1A). The redesigned SuperFi-Cas9 can edit DNA as efficiently as the natural Cas9 enzyme while being able to reduce the off-target probability by approximately 4000-fold. In practice, crRNA and tracrRNA are often fused into sgRNA to facilitate the operation of genetic engineering. Recently, Liu et al.[20] revealed that the programmability of crRNA-tracrRNA hybridization sequence could redefine the functions of Cas9 and diversify the sources of crRNAs. Particularly, the re-engineering tracrRNA pairs could recruit the endogenous small RNAs/mRNAs as crRNAs to activate the CRISPR/Cas9 system (Figure 1B). In addition, in practical applications, gene editing operations on selected target genes can be achieved only by replacing 20 nucleotide sequences of different target genes in the gRNA. However, in the structural study of gRNA, it was found that the misfolding of gRNA may directly reduce the editing efficiency, or the editing efficiency may be reduced by the misfolded gRNA competing with the active gRNA to bind the Cas9 enzyme. Moreover, the presence of TT and GCC motifs in the four bases proximal to the PAM in the spacer sequence was found to increase the possibility of sgRNA inefficiency.[21] Recently, Riesenberg et al.[22] found that modified tracrRNA could increase the cleavage efficiency of Cas9 protein by introducing highly stable hairpins and chemical modifications. The first hairpin of the tracrRNA with an extremely stable C(UUCG)G loop motif provides an efficient nucleation site for RNA folding to prevent misfolding of the tracrRNA, and chemical modification of tracrRNA can improve its stability against

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Table 1. Screening Markers Applied in Filamentous Fungi and Macrofungi

classification	screening marker	characterization or screening conditions	application species
Nutritional markers	*amdS*	The strain with *amdS* gene can grow normally on the medium with acetamide as the sole nitrogen source.	*A. niger*,[32] *M. thermophila*[33] et al.
	niaD	The *niaD* function-deficient mutant strain can tolerate high concentrations of chlorate.	*A. oryzae*[34] et al.
	pyrG	Mutant strains with *pyrG ura3* functional deletion can only grow normally on medium supplemented with uridine or uracil.	*A. niger*,[29] *P. eryngii*[28] et al.
	ppt1	Mutants with loss of *ppt1* function can only grow normally on medium supplemented with lysine.	*F. fujikuroi*[35] et al.
Resistance markers	Hygromycin (hyg^r)	*Hph* (Encoding hygromycin phosphotransferase) can inactivate hygromycin B.	*A. niger*,[29] *P. ostreatus*[36] et al.
	phosphinothricin (bar^r)	*Bar* (encoded product acetylated phosphinothricin) can inactivate phosphinothricin.	*G. lucidum*,[37] *M. thermophila*[38] et al.
	carboxin (cbx^r)	Point-mutated *sdhB* (encodes succinate dehydrogenase) in *G. lucidum* can inactivate carboxin.	*L. edodes*,[39] *G. lucidum*[37] et al.
	Neomycin (neo^r)	*Neo* (encodes aminoglycoside phosphotransferase) can inactivate G418 (neomycin derivative).	*P. sojae*,[40] *M. thermophila*[33] et al.
	Pyrithiamine (*ptrA*)	*PtrA* (point mutation in the gene encoding thiamine thiazole synthase) can inactivate pyridinethiamine.[41]	*A. oryzae*[34] et al.
Phenotypic reporters	*fcc1*	Disruption of the *fcc1* gene allows the accumulation of specific purple pigments.	*F. fujikuroi*[35]
	wA	The *wA* mutant forms white conidia.	*A. oryzae*[34]
	yA	The *yA* mutant forms yellow conidia.	*A. oryzae*[34]
	fwnA	The *fwnA* mutant forms white conidia.	*A. niger*[42]
	albA	The *albA* mutant forms white conidia.	*A. niger*[32]
	pkaC	Disruption of *pkaC*, which encodes the catalytic subunit of a cAMP-dependent protein kinase, resulted in a 2- to 3-fold reduction in colony diameter on CM plates.	*A. niger*[32]
	pksP	*PksP* knockout colonies had a distinct aconidial albino phenotype.	*A. fumigatus*[43]
	CnaA	Dysfunction of *CnaA* results in significant mycelial growth defects with very small and dense colony phenotypes.	*A. fumigatus*[43]
	cbh1	Disruption of *cbh1* will result in the loss of major bands on SDS-PAGE gels, thus facilitating the identification of strains by successful genome editing.	*T. reesei*[44]

cellular nuclease degradation through protecting the nucleotides at the 3′ and 5′ ends of tracrRNA using phosphonothioate bonds and through replacing the 2-hydroxyl group of the nucleotide with a methoxy group. These chemical modifications were combined with hairpin locking to obtain the "Genome editing Optimized Locked Design" (GOLD)-gRNA (Figure 1E). When using previously reported spacer sequences of inactive gRNAs with cleavage repressor motifs, such as TT- and GCC-motifs in the 4 PAM-proximal bases of the targeting sequence,[21] GOLD-tracrRNA improved editing efficiency by 6.8-fold compared to regular tracrRNA, indicating that this optimized design can perform robust editing regardless of how the spacer sequence is composed.

These optimized designs of the CRISPR/Cas9 system demonstrate the unlimited potential of the CRISPR/Cas9 system. Although these optimized systems have not yet been applied in fungi, it is foreseeable that more efficient and precise gene editing operations can be achieved in fungi using the upgraded and optimized CRISPR/Cas9 in the future.

2. THE FUNCTIONAL COMPONENTS OF THE CRISPR/CAS9 SYSTEM

Due to the advantages of the CRISPR/Cas9 system, it has been widely used in recent years for gene editing of fungi, especially filamentous fungi and macrofungi. In this process, the CRISPR/Cas9 system not only inherits the advantages of traditional CRISPR systems in other species but also gradually develops a set of CRISPR systems suitable for filamentous fungi and macrofungi. The selection of the appropriate screening marker, promoter, expression method, and delivery strategy of Cas9 and gRNA is the key to the successful application of the system. In this section, we will focus on the above-mentioned key factors of the CRISPR system in filamentous fungi and macrofungi.

2.1. Screening Markers. The paucity of selected markers in filamentous and macrofungi is one of the major hurdles for mutant screening. The commonly used screening markers are phenotypic reporters, nutritional markers, and resistance markers. Phenotypic reporters are generally derived from unique phenotypes in specific species such as some pigment synthesis genes and lethal genes, which are generally not universal. The selected markers currently applied in macrofungi and filamentous fungi are shown in Table 1.

The insertion or knockout of a nutrient gene in some specific strain will allow the transformants to grow normally on a medium containing a specific nitrogen or carbon source to achieve the purpose of the screening with nutritional markers. The most widely used auxotrophic marker is the *pyrG* auxotrophic gene, which originates from the *ura3* gene used in the biosynthesis of uracil nucleotides in yeast to encode orotidine-5′-phosphate decarboxylase.[23] Since the uracil biosynthetic pathway is well conserved in fungi, the *pyrG/pyr4* gene in *Aspergillus niger* and the *pyrF* gene in *Aspergillus oryzae* are highly homologous to the yeast *ura3* gene. Knockout of these genes can inactivate orotidine-5′-phosphate decarboxylase, leading to a uridine or uracil auxotrophic strain that can only grow on medium supplemented with uridine or uracil. In addition, wild type orotidine-5′-phosphate decarboxylase is able to utilize 5-fluorooric acid (5-FOA) to produce the toxic substance 5-fluorouridine. Thus, only *ura3/pyrG* mutant strains can grow in the medium supplemented with uracil and 5-FOA, while wild-type strains cannot (Figure 2A). Using the technique of 5-FOA counter-screening, Dong et al.[24] modified

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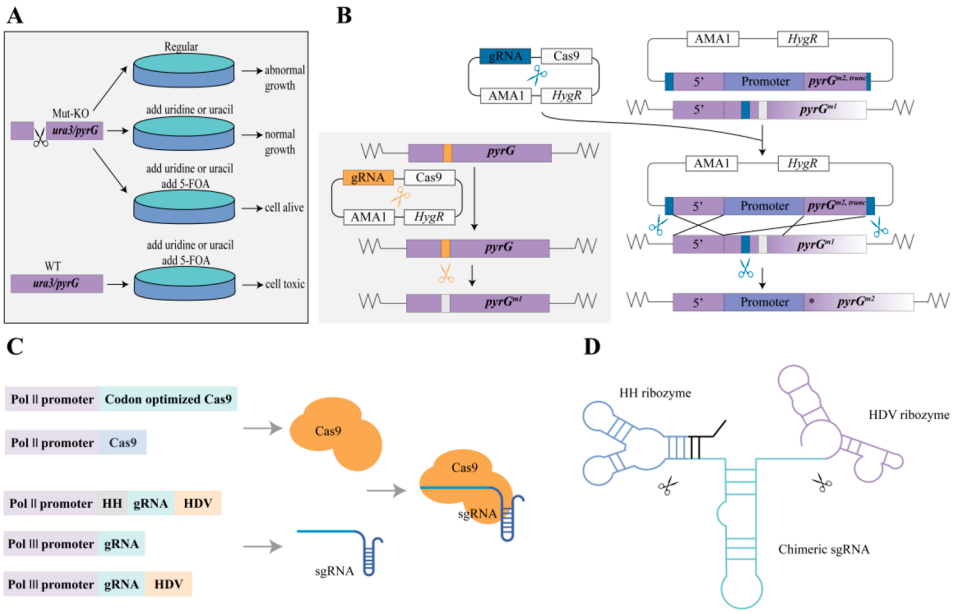

Figure 2. Recyclable selected marker technology and promoter selection of CRISPR-Cas9 system. (A) Growth of wild type strains with the *ura3/pyrG* gene and mutants with the *ura3/pyrG* gene knockout in different media. (B) A transient Cas9 expression plasmid containing the gRNA sequence for the *pyrG1* site was transformed into the parental strain *A. niger* to generate a double strand break at the *pyrG1* site (orange box). A strain *pyrGm1* carrying a mutation that leads to an auxotrophy for uridine is selected (mutation indicated as gray box). The integrated plasmid is cotransformed with the Cas9 plasmid containing the gRNA targeting the second site of the *pyrG* gene, namely *pyrG2* (blue box). The *pyrG2* site is also located on both sides of the integrated plasmid. Cas9 cleaves genomic DNA and releases linear integration boxes. The 5′ end of the cassette is homologous to the 5′ end sequence of the promoter region of the *pyrG* gene, while the 3′ end is the mutant and truncated *pyrG* CDS (*pyrGm2, trunk*), which has the mutant *pyrG2* site (*). In the *pyrGm2* strain, the *pyrG1* locus was restored. (C) For *in vivo* expression of Cas9 protein, strategies include codon-optimization, adopting constitutive promoter for driving the expression of Cas9 protein. For *in vivo* expression of gRNA, strategies include adopting a polymerase II promoter together with the self-cleaving ribozymes for gRNA expression, adopting a polymerase III promoter for gRNA expression, and adopting a polymerase III promoter together with the self-cleaving ribozyme for gRNA expression. (D) Liberation of the sgRNA, represented in light green, from a polymerase II transcript by intrinsic hammerhead (HH) ribozyme and hepatitis delta virus ribozyme (HDV) represented in blue/purple and orange, respectively. Cleavage points are indicated by scissors.

the *A. niger* CRISPR/Cas9 editing system by recycling and removing the autonomously replicating plasmids, which carry the marker genes pyrG and Cas9 expression elements.

However, the difficulty in obtaining auxotrophic strains of macrofungi, especially for some nonmodel fungi with relatively rare research, limits the application of this screening marker in genetic transformation. Moreover, the auxotrophic strains produced by mutation are unstable and might revert to wild type.[25,26] If the host strain does not contain the appropriate mutant allele required for the use of a prototrophic or an autoselection marker, a resistance marker needs to be employed.[27] Among the commonly used drug resistance markers, the hygromycin resistance gene is the most widely used, but it is also prone to the loss of exogenous resistance genes in practical applications. For example, Wang et al.[28] attempted to use hygromycin as a screening marker and established a CRISPR system in *Pleurotus eryngii*. However, the hygromycin resistance gene was lost after two selection passages, which might be caused by the relatively strong host defense mechanism of mushrooms, which causes the unstable integration of exogenous resistance genes in the genome of *P. eryngii*.

Due to the scarcity of selectable markers, researchers usually take various strategies to recycle marker genes or even use marker-free editing methods to expand the range of editable genes. For example, Sarkari et al.[29] constructed an ingenious CRISPR/Cas9 system in *A. niger*. They first used the CRISPR system to produce a *pyrG* insertion/deletion (indels) mutant strain (*pyrGm1*), allowing to perform uridine auxotrophy screening (Figure 2B). Then two separate plasmids, the integration plasmid and the Cas9/gRNA plasmid, were cotransformed into the above-mentioned *pyrG* mutant strain. The integration plasmid carried the target expression cassette, which is flanked by the promoter region of *pyrG* on the 5′ flank, a truncated mutated *pyrG* gene (*pyrGm1*) on the 3′ flank, and two synthetic gRNA sites *pyrG2* at two flanking ends. The Cas9/gRNA plasmid *in vivo* is responsible for cleaving the *pyrG2* sites located on both genomic DNA and the integration plasmid. The resultant broken genomic DNA and linearized integration cassette can yield the *pyrGm2* locus with the integrated cassette and restore the prototrophy by DNA double crossover (Figure 2B). Further, Leisen et al.[30] developed a strategy for cotransformaton of telomere-containing plasmids and Cas9-RNPs with or without donor template, to achieve marker-free CRISPR/Cas editing. The basic principle is that plasmids containing centromere (CEN4), replication initiation sequence (ARS1), and a pair of telomeres (pTEL) and carrying the selective marker Fen^r not only can be efficiently transformed into filamentous fungi and replicate here autonomously as centromere-free mini-chromosomes but also can be quickly lost in the absence of selective pressure. Using this approach, hundreds of transformants can be obtained repeatedly, of which with up to 65% are marker-free editing. In addition, Yuan et al.[31] developed a novel technique in plants that allows researchers to detect the activity of CRISPR gene editing tools in organisms using the

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Table 2. Promoters Applied to Filamentous Fungi and Macrofungi

type	name	source	applications
RNA polymerase type II promoter (constitutive promoter)	*tef1*	Yeast transcription elongation factor promoter	*A. niger*,[54] *M. thermophila*[33] et al.
	gpdA	The promoter of 3-phosphoglyceraldehyde dehydrogenase	*F. fujikuroi*,[35] *G. lucidum*[37] et al.
	pdc	The promoter of pyruvate decarboxylase	*T. reesei*[48] et al.
	trpC	Indolylglycerol phosphate synthase promoter	*Sclerotinia sclerotiorum*[55] et al.
RNA polymerase type II promoter (inducible promoter)	*cbh1*	The promoter of cellobiose hydrolase I	*T. reesei*[48] et al.
	glaA	The promoter of α-glucoamylase	*A. niger*[32] et al.
	xylP	Xylanase promoter	*A. fumigatus*[41] et al.
RNA polymerase type III promoter	*u6*	Human *U6* micronucleus promoter	*A. oryzae*,[34] *P. eryngii*[28] et al.
	5S rRNA	*A. niger* 5S rRNA gene	*A. niger*[32] et al.
	tRNA	Transcriptional transfer ribonucleic acid promoter	*A. niger*[51] et al.
In vitro transcription	*T7*	*T7* phage-derived promoter	*T. reesei*,[44] *P. chrysogenum*[56] et al.

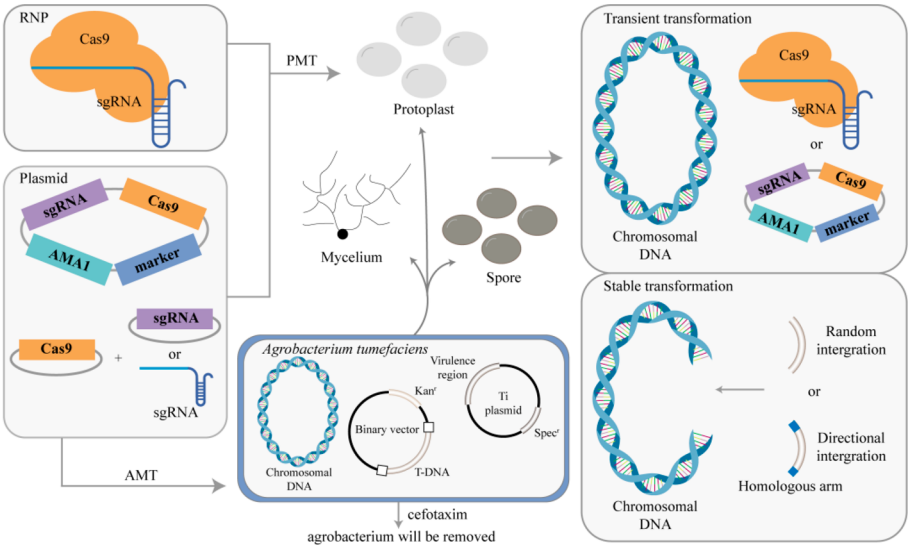

Figure 3. Expression and delivery strategies of CRISPR-Cas9. *In vitro* assembled RNPs, or plasmids carrying Cas9, gRNA, or the autonomous replication element AMA1, are introduced into host cells through AMT or PMT. The receptor for PMT can only be protoplasts, while AMT can be spores, mycelium, protoplasts. If RNP or the plasmid carrying AMA1 is introduced, CRISPR/Cas9 system is transiently expressed in the host cell. While ordinary plasmid is introduced, it is judged to be directional integration or random integration into the chromosome according to whether there is a homologous arm.

naked eye and a UV flashlight, and these biosensors may be an important tool for future mutant screening in fungi.

2.2. Selection of Promoters. Since the CRISPR/Cas9 system is an exogenous system, it needs to rely on endogenous promoters or some strong promoters for normal expression of exogenous genes *in vivo*. When applying the CRISPR/Cas9 system for gene manipulation, it is particularly important to select appropriate promoters for the expression of cas9 proteins, sgRNAs, and selected marker genes, respectively.

Generally there are three different promoters that bind specifically to the polymerase to mediate transcription initiation in eukaryotic cells. Among them, the RNA polymerase type I promoter can catalyze the synthesis of rRNA precursors and the RNA polymerase type II promoter is responsible for the transcription of mRNA, which is usually followed by a cap at the 5′ end of the transcript and a PolyA tail at the 3′ end to guide the transcript into the cytoplasm for protein translation. Type II promoters can be classified as constitutive promoters, tissue-specific promoters, and inducible promoters. And the RNA polymerase type III promoter mainly expresses noncoding short sequence RNAs, including sgRNA and shRNA (Figure 2C).

In fungi, the promoters for *in vivo* expression of gRNA are generally RNA polymerase type III promoters, such as *U6*, 5srRNA, and tRNA promoters. Among them, the *U6* promoter is one of the most commonly used promoters, especially in macrofungi. Under the control of *U6* promoter in *G. lucidum*, hepatitis D virus (HDV) self-cleaving ribozyme fused to the 3′ terminal of sgRNA can remove the terminal T of transcribed sgRNA and improve the gene editing efficiency[45] (Figure 2D). Generally, the *U6* promoter requires guanosine nucleotides to initiate transcription, thus limiting the sgRNA spacer sequence to GN19NGG, which reduces the number of available CRISPR/Cas target sites.[46,47] In addition, the low conservation of *U6* RNA sequence makes it difficult to identify the *U6* promoter in many species, often resulting in inefficient gene editing if a heterologous *U6* promoter is used.[48] For example, Zheng et al.[49] established a CRISPR/Cas9 system in *Nodulisporium* sp. (No. 65−12−7−1), using a U6 promoter derived from *A. oryzae*, but the genome editing efficiency was only 16%. As a latecomer among type III promoters, the 5S rRNA promoter has shown amazing efficiency. When using the endogenous 5S rRNA promoter in the CRISPR/Cas9 system of *A. niger*, its gene editing efficiency is nearly 96%, which is

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apparently higher than that of the previous *PhU6* or *PanU6* promoter systems.[32] In addition, tRNA promoters can also act as promoters for sgRNA to drive expression. Song et al.[50] tested the drive efficiency of 37 tRNA promoters in *A. niger*, of which 36 could successfully achieve gene editing. Subsequently, Kun et al.[51] used the $tRNA^{Pro1}$ promoter to drive sgRNA for homologous recombination editing in *A. niger* with up to 80% editing efficiency. The single gene disruption efficiencies of 92% and higher were achieved in *Yarrowia lipolytica* when the Pol III-tRNA hybrid promoters exploit endogenous tRNA processing to produce mature sgRNA for Cas9 targeting.[52] Attempts have also been made in filamentous fungi to use RNA polymerase type II promoters to express gRNA, usually by introducing a hammerhead nuclease (HH) at the 5′ end and a hepatitis D virus (HDV) nuclease at the 3′ end of the gRNA, both of which allow the transcript self-splicing to remove modifications at both ends, avoiding failure to perform guidance and targeting functions[53] (Figure 2D). In species where a validated type III promoter cannot be identified from the genome, *in vitro* transcription of gRNA with T7 Transcription Kit is also a proven option. This usually requires the construction of a Cas9 overexpression strain and the introduction of *in vitro* transcribed gRNA into the recipient strain for gene editing.

The promoter used to express heterologous proteins (cas9 or marker genes) is usually an RNA polymerase type II promoter, and the choice to use an inducible promoter or a constitutive promoter depends on whether regulated expression is required. In general, constitutive promoters are the most widely used, while inducible expression of Cas9 has many applications in filamentous fungi for its ability to regulate the viability of the CRISPR/Cas9 system during gene editing, reduce off-target effects, etc. The type III promoters and type II promoters commonly used in the CRISPR system are shown in Table 2.

2.3. Expression of the CRISPR/Cas9 System. CRISPR/Cas9 systems are typically expressed in three different ways. The first is to transform plasmids harboring Cas9, gRNA cassettes and appropriate promoters into cells.[57] The second is to transform the *in vitro* transcribed gRNA or gRNA-expressing vector into a Cas9 expression strain. The third one is to assemble ribonucleoproteins (RNPs) *in vitro* and then transform them into cells.[58] This section will discuss the related progress of these three expression modes of the CRISPR/Cas9 system, according to whether cas9 is expressed *in vivo* or *in vitro* (Figure 3).

2.3.1. In Vivo Expression of Cas9. *In vivo* expression of the CRISPR/Cas9 system is the most used and well-studied method in fungi so far. Although it is convenient to transfer Cas9 and gRNA in the same vector into host cells, sometimes the efficiency of system delivery and expression can be significantly reduced due to the large size of the vector. Therefore, the introduction of *in vitro* transcribed gRNA or gRNA-expressing vectors into Cas9 overexpression strains is also an option. Compared with transcription of gRNA *in vivo*, transcription of gRNA *in vitro* is not dependent on endogenous promoters but has limitations such as high cost and susceptibility to degradation by intracellular RNA enzymes.

Regardless of whether sgRNAs are transferred into the host cell together with cas9, the impact of continuous expression of cas9 *in vivo* on gene editing has both advantages and disadvantages. On one hand, a CRISPR-Cas9 system that is continuously expressed *in vivo* can increase the probability of targeting the target DNA, thereby improving the efficiency of gene editing, and delivery of sgRNA in strains expressing Cas9 protein can reduce the risk of Cas9 protein and sgRNA failing to enter the same nucleus. On the other hand, due to the use of Cas9 protein derived from the prokaryotic *Streptococcus pyogenes*, the coding genes in most cases require codon optimization for eukaryotic cells, hindering the implementation of CRISPR-Cas9 systems in organisms that lack databases for genome sequence. After Cas9-encoding gene is transformed into host strains using common delivery strategies, the *cas9* gene is generally randomly integrated into the genome, which may affect the host genome structure, leading to unexpected phenotypes, such as growth delay,[59] further rearrangements or off-target mutations,[60] and toxic effects on the host.[61] Therefore, it is important to ensure that Cas9 protein has no adverse effects on filamentous fungi by testing the biological properties, such as development, stress resistance and virulence of Cas9-expressing strains.[62−64]

In addition to improving the expression level of Cas9 by codon optimization, other strategies, including modifying Cas9 by nuclear localization signals to increase the entry of Cas9 protein into the cell nucleus, or increasing the expression level of Cas9 by appropriate introns, can also improve editing efficiency. For example, Shi et al.[35] successfully established CRISPR/Cas9-mediated gene editing technology in the filamentous fungus *Fusarium fujikuroi*, and found that the nuclear localization signal from histone H2B (HTB_{NLS}) could guide Cas9 proteins into the nucleus more efficiently by comparing three different nuclear localization signals ($SV40_{NLS}$, VEL_{NLS}, HTB_{NLS}). In addition, some researchers have also placed SV40 nuclear localization signal (PKKKRKV) and nucleoplasmin nuclear localization signal (KRPAATKKAGQAKKKK) at the N-terminal and C-terminal ends of Cas9, respectively, to enhance the nucleation efficiency.[32] While Liu et al.[37] used the macro fungus *G. lucidum* as a host and found that introns were essential for efficient expression of exogenous genes (glufosinate resistance and green fluorescent protein genes). Based on this discovery, they constructed an engineered *G. lucidum* strain highly expressing Cas9 protein by adding an intron to the 5′ end of the *cas9* gene. In this engineered strain, the disruption efficiency of gene *ura3* was 10.6-fold higher than that of the previously reported strain without an extra intron in the *cas9* gene.

The transient expression of Cas9 protein by introducing non-replicating plasmids into protoplasts can reduce the probability of gene integration into chromosomes and reduce the potential toxic side effects caused by gene integration disrupting the genome.[65] However, with this approach, the transient plasmids can also be degraded intracellularly by endogenous nucleases, and the resulting small DNA fragments may still be inserted into target and off-target sites in the host cell.[66] The AMA1 autonomous replication element found in *A. niger* was used to construct a Cas9 expression plasmid, which could enter the cell and replicate itself without integrating exogenous genes into the host genome.[29] On the basis of the AMA1 plasmid, Aoace2 gene cassette was inserted under the control of *PamyB* promoter, and the dextrin-induced Aoace2 expression could severely inhibit cell growth, which make this plasmid more susceptible to loss, leading to the plasmid recycling and repeated genome editing.[34]

2.3.2. In Vitro Assembly of CRISPR/Cas RNPs. Unlike plasmid vectors that deliver Cas proteins and gRNAs into cells for *in vivo* expression, CRISPR/Cas ribonucleoproteins

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(RNPs)-mediated genome editing technology does not have the risk of integrating foreign DNA into the genome during genome editing. This technique preassembles purified Cas proteins and gRNAs into RNPs with complete activity *in vitro* and then introduces RNPs directly into cells by physical or chemical methods, avoiding the insertion of exogenous DNA into the recipient genome at source.

Therefore, this approach has distinct advantages. First, the assembly of Cas9-gRNA RNPs is not limited by the amount or rate of Cas9 translation or gRNA transcription and protects gRNA from degradation. Furthermore, genome editing by delivery of RNPs reduces unwanted gene targeting and avoids integration mutagenesis that may occur as a result of gene delivery strategies.[58,67] At the same time, because RNPs with complete activity are introduced, RNPs can function immediately after entering the cell and rapidly cut the genomic DNA of the target site. Due to the relatively short half-life of Cas proteins and guide RNAs, after the RNP complex has completed cleavage of the target site, the cell initiates repair mechanisms and resumes growth and metabolic processes, at which time the RNP complex is naturally degraded *in vivo*, thus achieving nearly "traceless editing". Currently, the application of gene editing using *in vitro* synthesis of RNP has been achieved in filamentous fungus *M. oryzae*[68] and in macrofungi *P. ostreatus*[36] and *S. commune*[69] (Table 3). In addition, studies in many species, including *P. ostreatus*,[36,70] have shown that an appropriate increase in the molarity of RNPs transformation can increase the number of positive transformants to a certain extent.

From a cost perspective, the cost of *in vitro* transcription of gRNA is relatively high and the RNP complex is easily degraded during the transformation process, which puts forward higher requirements for the genetic transformation operation. Moreover, screening mutants is more time-consuming and labor-intensive due to the lack of resistance markers. One study showed that using the classical protoplast transformation method to transform RNP into *T. reesei*, the editing efficiency was extremely low, with only 5 of 143 transformants repaired by homologous recombination.[44] These may be one of the reasons there are not many related studies in fungi at present. In addition to this, it remains a great challenge to efficiently transport the assembled RNP *in vitro* to the cell and then localize it to the nucleus. Since fungal protoplasts are relatively small, the probability of delivering RNPs to protoplasts using PEG-mediated protoplast transformation is also extremely low, which also leads to low editing efficiency of RNPs.[71]

To address these issues, Zou et al.[70] proposed an optimization strategy to improve the efficiency of genome editing using RNPs in filamentous fungi by adding chemical reagents. They used the surfactant Triton X-100 to improve cell membrane permeability and increase the probability of RNPs entering the protoplasts. In *T. reesei*, the optimized RNP-based method also achieved effective (56.52%) homologous recombination integration using a short homology arm (20 bp) and gene disruption (7.37%), excluding any exogenous DNA (selection marker). And they also used inositol and benomyl for controlling cell division and mitosis, by which they successfully increased the transformation efficiency of multinucleate spores and protoplasts of *A. oryzae* for homokaryotic transformation (from 0% to 40.0% for inositol and 71.43% for benomyl). In this study, it was possible to establish efficient RNPs-based CRISPR/Cas9 systems in *T. reesei*, *C. militaris*, and *A. oryzae* with these chemical reagents, which demonstrated the good versatility of the method in fungi. And the strategy of using nanoparticles to deliver cas9 protein, which was successfully applied on animal cells in earlier years,[66,72] has recently been successfully applied in fungi. Li et al.[68] utilized calcium phosphate-mineralized nanoparticles to carry RNPs for delivery into protoplasts of the plant pathogenic fungus *M. oryzae*, in which biomimetic mineralized RNPs protected gRNA from enzymatic hydrolysis, and preserved the enzymatic activity of Cas9 protein.

2.4. Delivery of the CRISPR/Cas9 System. Currently, two commonly used CRISPR/Cas9 system delivery methods are poly(ethylene glycol)-mediated transformation (PMT) and *Agrobacterium tumefaciens*-mediated transformation (AMT). The CRISPR/Cas system delivered by PMT can be DNA, RNA, or RNPs assembled *in vitro*[73] (Figure 3). PMT is a transformation method based on protoplast preparation. Protoplasts are mixed with exogenous DNA in polyethylene glycol (PEG) buffer containing $CaC1_2$, and the exogenous DNA enters the host cell after the permeability of host cell membrane changes; then recombinant strains are obtained through protoplast regeneration and transformant screening. Notably, exogenous DNA fragments are usually randomly integrated into the filamentous fungal genome in the form of multiple copies after PEG-mediated transformation.[74] In this mode of delivery, the preparation and regeneration of protoplasts are critical steps. The selection of appropriate tissue material (spores, nascent mycelium, etc.), growth phase of the strain, suitable cell wall digesting enzymes, certain concentrations of osmotic pressure stabilizers, and matching regeneration medium are crucial for the successful preparation of protoplasts. Therefore, the PMT method does not have a certain simplicity in terms of operating cost and process and is more difficult to achieve gene editing for some fungi. The protoplast method was first applied to *Neurospora crassa* in 1958,[75] and currently, the PEG-mediated protoplast delivery CRISPR/Cas9 system has been successfully applied to a series of fungi, including *A. niger*,[76] *A. oryzae*,[34] *P. eryngii*.[28] *Agrobacterium tumefaciens* can infect any higher organism and transfer a DNA fragment (transfer DNA, T-DNA) from a tumor-inducing (Ti) plasmid into the host genome via activity of *vir* gene product (Figure 3). This DNA-transferring capacity of *A. tumefaciens* is used for the transformation of fungi. For this purpose, a so-called binary vector system is used, meaning that *A. tumefaciens* contains two plasmids besides its chromosome. A binary vector contains the T-DNA, and the Ti plasmid is devoid of its natural T-DNA, but it contains *vir* genes that encode for the T-DNA transfer machinery. There is a 24 bp repeat sequence on the left and right borders (LB; RB) of T-DNA in the binary vector, which is the *cis*-acting signal of the DNA delivery system. After induction of the T-DNA transfer machinery, any DNA placed between these border repeats is transferred into the host as a single-stranded DNA molecule, where it will randomly integrate into the host genome. After transformation, *A. tumefaciens* will be removed from the culture by using cefotaxim as selection pressure.[77] Using this property, AMT has been applied to the transformation of a variety of fungi, including *C. militaris*,[78] *Trichoderma reesei*.[79] Compared with PMT, a wide variety of receptor, such as fresh spores, conidiospores, mycelia, and protoplasts, can be subjected to AMT. Another major advantage of AMT is that T-DNA is an efficient substrate for homologous recombination when sequences homologous

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Table 3. Application of CRISPR/Cas9 System in Typical Species of Macrofungi and Filamentous Fungi

classification	species	expression of gRNA	expression of Cas9	materials and delivery methods	editing methods	target gene and selected marker	mutation efficiency	reference
Filamentous fungi	*A. fumigatus*	T7 *in vitro* transcription/U6 expression *in vivo*	P*gpdA*/P*niiA*, T*trpC*, human codon-optimized hCas9	Protoplast, PMT	NHEJ/ HDR	*pksP, CnaA; pyr4, hph, GFP*	95−100% mutation rate	2016[43]
	A. niger	Type II promoter p*mbfA*, T*trpC*	P*coxA*, T*tef1*	Protoplast, PMT (AMA1)	NHEJ/ HDR	*pyrG; hygr*	2 transformants with the correct integrated form were obtained for each 1 μg of DNA transformed (out of 7 transformants)	2017[29]
		Endogenous 5S rRNA promoter	P*glaA*, T*glaA*, codon-optimized NLS-Cas9-NLS	Protoplast, PMT	HDR	*Alba, pkaC, fumS, fum1; hph, amds*	33.3−100% mutation rate	2018[32]
		Endogenous Pan*U6* promoter, exogenous Ph*U6* and PyU6 promoter	P*glaA*, T*glaA*, codon-optimized Cas9	Protoplast, PMT	HDR	*albA; hph, amdS*	Pan*U6*: 79%	2018[110]
		U6 promoter from *A. oryzae*	P*tef*, T*tef*, Codon-optimized Cas9 fused to APOBEC1 to form BEC	Protoplast, PMT (AMA1)	BE	*pyrG, fwnA, prtT (nonphenotypic) hygr, pyrG*	Conversion of cytosine to thymine at specific locations in target genes with 47.36% to 100% efficiency	2019[42]
		Af*U6*, Ao*U6*; T7 *in vitro* transcription	pFC330 (*tef1*)	Protoplast, PMT (AMA1)	HDR	*fwnA, amyA, glaA; hygB, amdS*	*amyA*: 57%; *glaA*: 100%, 57% mutation rate for simultaneous knock-in at two sites	2019[24] 2020[54]
		$tRNA^{Pro1}$ promoter	P*pkiA*, T*glaA*, Cas9(ANEp8-Cas9-*pyrG*)	Protoplast, PMT (AMA1)	HDR	*Xln R, GaaR; pyrG*	80% mutation rate	2020[51]
	A. oryzae	*U6* promoter	PAo*tef1*, P*amyB*	Protoplast, PMT (AMA1)	NHEJ/ HDR	*wA, pyrG, yA, niaD; ptrA, GFP*	55.6−100% mutation rate	2019[34]
	F. fujikuroi	Endogenous *U6* or 5SrRNA promoter	P*gpdA* and T*trpC* from *A. nidulans*, codon-optimized Cas9	Protoplast, PMT	NHEJ	*fcc1, ura3, ppt1; hph*	*U6*: 37.5%; 5SrRNA: 79.2%	2019[35]
	M. oryzae	*In vitro* synthesis	*In vitro* synthesis	Coculture of RNPs with protoplasts using calcium phosphate-mineralized nanoparticles	NHEJ	*sdh*	The efficiency of interrupting *sdh* gene is 20%	2019[68]
	M. thermophila	Endogenous *U6* promoter, 6T terminator	Endogenous P*tef1*, T*tprC*	Protoplast, ATMT	NHEJ/ HDR	*amdS/ (FAA), cre1, res1, gh1, alp; bar, neo*	Single Gene: 90−95%; Dual Gene 61−70%; Four Genes 22%	2017[33]
		Endogenous *U6* promoter, 6T terminator	Endogenous P*tef1*, codon-optimized Cas12a	Protoplast, PMT	NHEJ/ HDR	*amdS/(FAA), cre1, res1, gh1, alp; bar, neo*	Single Gene: 90%; When targeting multiple genes, single gene editing occurs with 22% to 41% efficiency	2019[38]
	P. chrysogenum	Endogenous *U6*, tRNA, Utp25 promoter or *in vitro* synthesis	P*xyl* or *in vitro* synthesis	Protoplast, PMT	HDR	*pks17, roqA, lovF, pcbAB, penDE, hcpA; amdS*	Up to 60−100% mutation rate	2016[71] 2018[56]
	T. reesei	T7 *in vitro* transcription	P*pdc*/P*cbh1*, codon-optimized Cas9	Protoplast, ATMT, PMT	HDR	*ura5, lae1, vib1, clr2; poura5, hyg*	Single Gene: up to 93−100%; simultaneous editing of multiple genes: 4.2−45%	2015[48]
		T7 *in vitro* transcription	*In vitro* synthesis	Protoplast, PMT	NHEJ	*cbh-1; pyr4*	40% mutation rate	2019[44]
		Endogenous *U6* promoter	P*pdc*/P*cbh1*, codon-optimized Cas9	Spore, ATMT (electroporation)	NHEJ	*ura5*	high efficiency	2020[79]
macrofungi	*A. bisporus*	Endogenous *U6* promoter	Endogenous *gpd* promoter	Mycelium, ATMT	NHEJ	*ppo4, hygr*	2 of the 12 transformants produced mutations	2020[111]
	C. cinerea	Endogenous *U6* promoter	Endogenous CcDED1 promoter, codon-optimized Cas9	Protoplast (cryopreservation), PMT	NHEJ	*Gfp, hygr*	10.5% mutation rate	2017[112]
	C. militaris	T7 *in vitro* transcription	Endogenous *gpd* promoter, codon-optimized Cas9	Spore, ATMT; Protoplast, PMT	NHEJ	*ura3, blpRr*	11% mutation rate	2018[78]
	G. lucidum	T7 *in vitro* transcription	Endogenous *gpd* promoter, codon-optimized Cas9	Protoplast, PMT	NHEJ	*ura3, cbxr (sdhB)*	0.2−1.78 (mutants/10^7 protoplas)	2017[113]
		T7 *in vitro* transcription	Endogenous *gpd* promoter, codon-optimized Cas9	Protoplast, PMT	NHEJ	*ura3/GL17624, barr, sdhB*	Single site: 28.5−34.6%; Two sites: 36.7%	2020[37]

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Table 3. continued

classification	species	expression of gRNA	expression of Cas9	materials and delivery methods	editing methods	target gene and selected marker	mutation efficiency	reference
		Endogenous *U6* promoter	Endogenous *gpd* promoter, codon-optimized glcas9	Protoplast, PMT	NHEJ/HDR	*ura3/cyp51S018/cyp50S413, cbxr*	0.67−5.3 (mutants/10^7 protoplas)	2020[45]
	P. eryngii	Endogenous *U6* promoter	Endogenous *gpd* promoter, codon-optimized Cas9	Protoplast, PMT	NHEJ/HDR	*pyrG, cbxr; hph, GFP*	78.9% mutation rate	2021[28]
	P. ostreatus	*In vitro* synthesis	*In vitro* synthesis	Protoplast, PMT	NHEJ/HDR	*pyrG, hygr*	7−100% mutation rate	2021[36]
		Endogenous *U6* promoter	*ef3* promoter from *C. cinerea*, codon-optimized Cas9	Protoplast, PMT	NHEJ/HDR	*fcy1/pyrG, hygr*	83.3% mutation rate	2021[114]
		Endogenous *U6* promoter	*ef3* promoter from *C. cinerea*, codon-optimized Cas9	Protoplast, PMT	NHEJ	*pcc1/clp1, hygr*	high efficiency	2021[115]
	S. bambusicola	*U6* promoter from human	Endogenous *ef1α* promoter, Cas9	Protoplast, PMT	HDR	*SbaPKS, hygr*	32% mutation rate	2017[116]
	S. commune	*in vitro* transcription	*in vitro* synthesis	Protoplast, PMT	HDR	*hom2, nourr*	0.15−1.8 (mutants/10^7 protoplas)	2019[69]

to the host genome are present in the T-DNA, leading to relatively high gene-replacement frequencies. Finally, as integration in fungi appears to occur predominately as a single copy, at random and in a nonsequence specific manner when there is no homology between the T-DNA and the host genome, this transformation system can insert large fragments of exogenous genes and can obtain transformants with stable genetic traits.[80]

The transformation methods can be divided into stable transformation and transient transformation according to the time when the foreign gene exists in the cell (Figure 3). In the stable transformation mode, exogenous genes are integrated into the cell chromosome and generally do not disappear with the passage of cells. In this transformation method, methods such as the linearization of plasmids are usually used to increase the probability of exogenous gene integration into the genome. Stable transformation is able to produce the target protein continuously for a long time, but also consequently increases the expression of cas9 protein and increases the probability of off-targeting.[60] Transient transformation is usually performed using a vector with an autonomously replicating element (AMA1) or using an *in vitro* assembled RNP as a CRISPR/Cas9 system for transformation. With this type of transformation, the exogenous gene is not integrated into the chromosome of the cell, but exists on a free vector, so that the gene expression product can be obtained for a short period of time, but the exogenous gene will eventually be lost as the cell continues to divide and proliferate. Transient transformation has many advantages, such as the ability to reduce selected markers or even perform markerless gene editing, and the ability to carry multiple copies of the host gene with high expression efficiency.

3. APPLICATIONS OF CRISPR/CAS9

In general, editing by the CRISPR/Cas9 system in fungi mainly involves the introduction of specific deletions and point mutations through CRISPR/Cas9 system by means of nonhomologous end joining (NHEJ) or homologous repair (HDR).[81] This section focuses on the application of the CRISPR/Cas9 system in filamentous fungi as well as macrofungi in recent years, respectively.

3.1. NHEJ-Mediated Gene Knockout. In fungi, DSBs generated by Cas9 in the absence of donor template DNA are repaired mainly by nonhomologous end joining (NHEJ), which joins the broken ends of DNA together and is a low-fidelity repair process. Random insertion or deletion of bases will occur in the process of NHEJ, resulting in frameshift mutations, thereby achieving gene inactivation or mutation of the target gene (Figure 4A).[82] Although NHEJ-mediated insertion or deletion is largely random, gene disruption can be ensured by targeting the gRNA to the N-terminus of the gene of interest.[83,84]

In the early application of the CRISPR/Cas9 system, the gene disruption of the target species was usually performed in an NHEJ-mediated method to reveal the possible function of the gene.[85] In addition, NHEJ-mediated knockout can also be used to interrupt endogenous marker genes in the genome to verify whether the CRISPR/Cas9 system was successfully constructed. However, NHEJ-mediated editing of many cell lines with a small number of insertions/deletions produces aberrant transcripts or proteins, leading to unintended effects that complicate functional analysis.[86] Although the indels mutation usually deprives the gene function, it cannot be ruled

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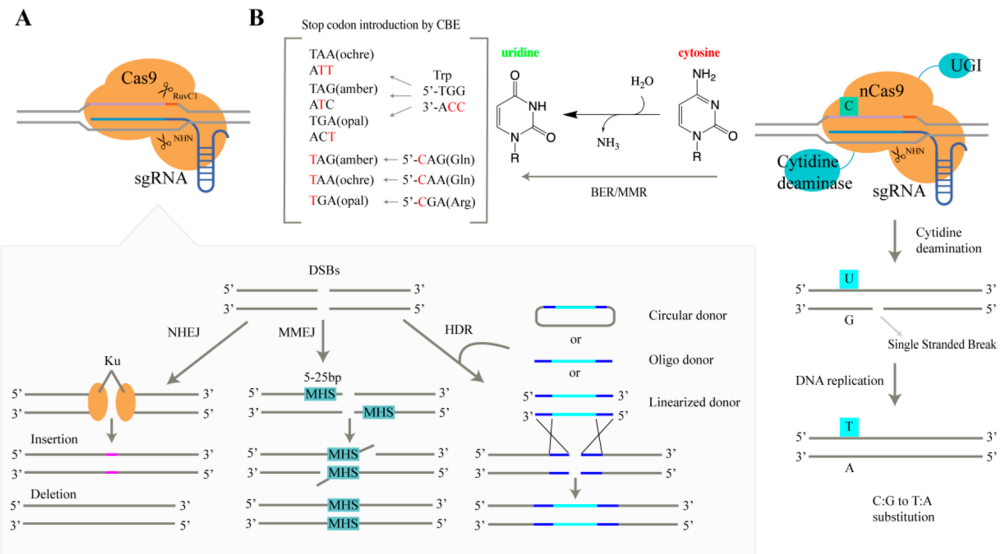

Figure 4. Three repair mechanisms of DSBs caused by Cas9 and the mechanism of CBE. (A) Repair mechanisms of NHEJ, MMEJ, and HDR. During NHEJ repair of a DSB, the Ku protein binds to the DSB and prevents DNA end resection, resulting in small deletion and insertion products. MMEJ requires end resection or unwinding to reveal homologous sequences. The length of homology required for MMEJ (5−25 bp) is short. The HDR repair pathway requires the involvement of donors with homologous arms. (B) The mechanism of CBE. CBE is composed of nCas9, UGI, and the deaminase domain. It uses BER to repair uracil into cytosine, which can then be recognized as T by polymerase. UGI is added to prevent the modified U from being removed by MMR. Due to the inability of the modified U to continue pairing with the unmodified G, DNA nicks (nicks the nonedited strand) were produced using nCas9 on the G-containing strand opposite the newly formed uracil to induce MMR in the host cell, cutting G into A paired with U.

out that mutations maintain the open reading frames. Therefore, NHEJ mediated editing can deprive these functions of genes but is not absolute. Instead, targeting two or more sites in a gene or chromosome region may result in the deletion of hundreds of bases between the targeted sites.[87] With this strategy, an independent exon of a gene or the entire gene can be knocked out.

Microhomology-mediated end joining (MMEJ) is another DNA repair mechanism that utilizes a 5−25 bp microhomologous sequences (MHSs) located near DSBs to mediate the repair of mismatch-prone end-joined DSBs[88] (Figure 4A). It was shown that MMEJ usually results in sequence deletions between pairs of MHS near DSBs generated by genome editing.[89,90] In plants, the efficiency of MMEJ-mediated deletion was as high as 52.4%.[91] Currently, there are few studies using this method for gene knockout in filamentous fungi and macrofungi, but after clarifying the mechanism of each species, it is also a feasible method to use MMEJ mediated genomic fragment deletion.

3.2. HDR-Mediated Gene Knock-in. If a DNA repair template (donor DNA) is provided, the CRISPR/Cas9 system enables homologous repair (HDR) in fungi, thereby introducing a specific point mutation at the target site, inserting the desired sequence, or replacing the sequence exactly with the desired sequence.[92] Numerous studies have shown that HDR-mediated editing is less efficient in NHEJ-active wild-type strains.[71,93] Since the repair mechanism of NHEJ in eukaryotes may be mediated by Ku70/Ku80 proteins, the efficiency of homologous recombination would be greatly enhanced after targeted mutation of *Ku70/Ku80* genes[80] (Figure 4A). For example, in 2010, Jong et al.[94] mutated the *ku80* gene in the mushroom *Schizophyllum commune*. Although the transformation efficiency of the mutated ku80 strain was reduced nearly 100 times, the relative number of transformants with homologous integration increased dramatically. Subsequently, Vonk et al.[69] found that the use of Cas9 RNP in *S. commune* strains with *Ku80* gene mutation improved the efficiency of homologous recombination and had a higher rate of gene deletion compared to the wild type.

Undoubtedly, NHEJ, simpler and faster than HDR, grants a cell more capacity to repair DSBs and safeguard its genome. Consequently, interfering with a critical DSB repair mechanism negatively impacts cellular fitness/growth, potentially hindering microbial factory capacity.[95] A clear strategy to obtain precise editing is to find alternate ways to improve NHEJ-independent HDR-mediated genome editing. The gene integration strategy for initiating HDR in some nontraditional yeast species with NHEJ dominated DNA repair is also worth learning. Deon et al.[96] showed a CRISPR platform that temporal control of Cas9 expression (fusion of Cas9 with geminin that can inhibit DNA replication) and repackaging the delivery of CRISPR expression components and donor DNA in a single linear DNA fragment boosted precise genome editing to efficiencies of 67−100% in several NHEJ-proficient strains. Peng et al.[97] established a CRISPR-Cas9 based genome editing platform in *Pichia pastoris*, which improved the HDR efficiency by more than 54 times, in particular, enhanced the simultaneously assembly of multiple fragments by 13.5 times. Specifically, they enhanced HDR process in *P. pastoris* by overexpressing the relevant enzymes. Then they enhanced the overexpression of endogenous structure-selective endonucleases (SSEs) that recognize and cleave the structures formed at the boundaries of DNA strand exchange intermediates and have key functions in the multi-invasion-induced rearrangement (MIR) process. Schwartz et al.[98] engineered a rapid and reliable method in *Yarrowia lipolytica* that has the capability to integrate multiple genes into different loci without marker recovery.

The size of the homologous arm is also critical to achieve efficient and accurate gene knock-in. Liu et al.[48] compared the

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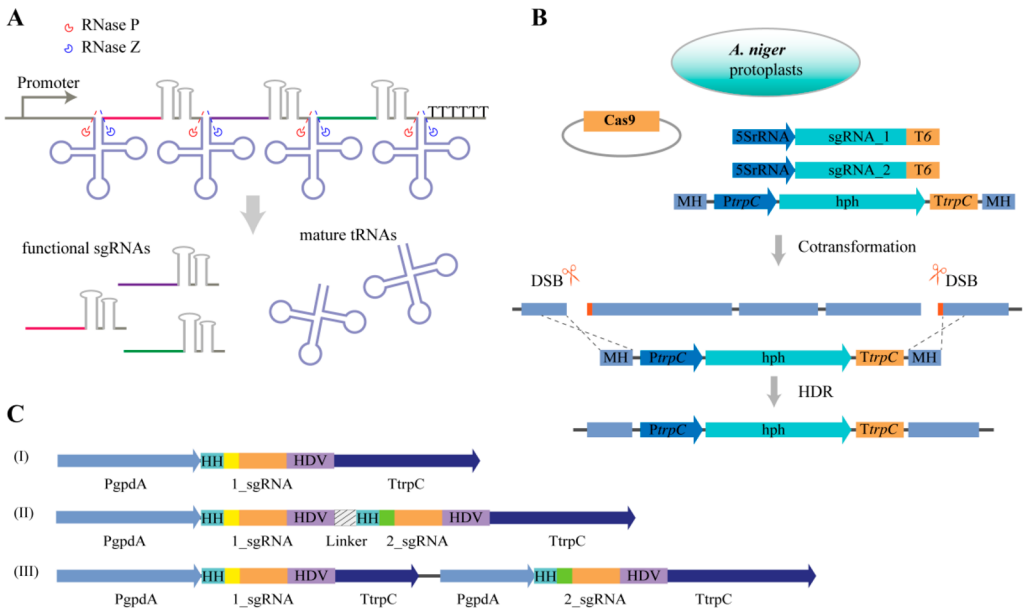

Figure 5. Multiplex editing strategies of CRISPR−Cas9 by simultaneously expressing multiple gRNAs. (A) The tRNA-gRNA array can be processed by endogenous RNase P and RNase Z to release functional sgRNAs. (B) sgRNAs with different protospacers and donor DNA with microhomology arms were cotransformed with a Cas9 expression plasmid into *A. niger* protoplasts. (C) Primary sgRNAs with flanking ribozyme sequences are self-cleaved to generate functional sgRNAs. (I) The construct contains only single target sgRNA cassette. (II) In the single-promoter, dual-targeting constructs, the 1_ and 2_ sgRNAs are each embedded within flanking ribozyme sequences, are separated by a linker (hatched bar), and share a single promoter and terminator region driving their expression. (III) Dual-promoter, dual-targeting constructs have separate promoter-sgRNA-terminator cassettes for each targeted gene.

effect of homologous arms of different lengths (0.2−1.0 kb) on the editing efficiency of the CRISPR/Cas9 system in *T. reesei*. The homologous recombination frequency of all tested homologous arms was ≥93%, indicating that a pair of 200 bp homologous arms was sufficient to achieve an effective homologous integration stimulated by the CRISPR/Cas9 system in *T. reesei*. Research on *A. niger* found that short homologous arms of no more than 40 bp can help CRISPR/Cas9 achieve precise genome editing. Zheng et al.[32] achieved 75% homologous recombination efficiency in a strain with an active NHEJ system of *A. niger* using a homology arm only 40 bp short, while precise homologous targeted repair with 100% editing efficiency was achieved in strains with disrupted NHEJ pathway. Specifically, construction of sgRNA with different protospacers and donor DNA with microhomology arms was followed by cotransformation of Cas9 expression plasmid, sgRNA, and donor DNA into *A. niger* protoplasts subjected thereafter to appropriate selective medium (Figure 5B). With this strategy, they successfully deleted the 48 kb gene cluster in the NHEJ deficient hosts with 100% efficiency. Besides, Dong et al.[24] discovered that a 39 bp short homologous arm can also help CRISPR/Cas9 achieve accurate genome editing of *A. niger* and the closer the homologous arm is to the DSB, the higher the efficiency of genome editing. It was shown that the homology arm was within 5 kb upstream and 5 kb downstream of the target gene PAM site to achieve deletion of the genomic fragment, but as the distance of the homology arm from the double-stranded gap increased, the editing efficiency decreased accordingly, with a maximum editing efficiency of 80% at 0 bp from the cut and a homology arm length of 39 bp.

It has also been recently shown that circular donor DNA can be integrated into target motifs more frequently than linear donor DNA, possibly due to the absence of free ends available for binding to NHEJ proteins and therefore more favorable for HDR repair.[99] Besides double-stranded DNA that can be used as a repair template, Nødvig et al.[100] found that single-stranded oligonucleotides can also be used as donors (Figure 4A). They successfully introduced nonsense codons into the pigmentation genes (yA, alba, and wA) of *A. niger* using 90-mer single-stranded oligonucleotides and also showed in their study that DNA repair has no preference for targeting the sense strand or the antisense strand. Kun et al.[51] investigated the editing efficiency of single-stranded oligonucleotides of different lengths on this basis and found that 60-mer or even shorter repair templates could also successfully repair damaged DNA.

A new method called "base editor" (BE) has been developed in mammalian cells to avoid DNA damage during gene editing and to eliminate the need to deliver HDR donor templates.[101] The single base editing system consists of two components, the cytosine base editor (CBE) and the sgRNA, which can convert cytidine (C) to uridine (U) at the target site without generating a DSB (Figure 4B), then U can be recognized as T by polymerase.[102] The editing window size for this base conversion is 5 nt, located at 3−8 nt of the protospacer. CBE is a tripartite fused protein, including a base modification enzyme, UGI (Uracil DNA glycolase inhibitor), and Cas9 nickase (nCas9) or dead Cas9 (dCas9). The base modification enzyme is a deaminase that catalyzes the formation of uracil from cytosine.[101] UGI is a structure that inhibits the activity of intracellular uracil glycosidase in the host cell to prevent modified U from being removed by the base mismatch repair (MMR) mechanism (Figure 4B). nCas9 is a variant of Cas9 that mutates the D10A amino acid in Cas9 causing single strand breakage and base excision repair (BER). dCas9 is also a variant of Cas9, which mutates the parts responsible for splicing in Cas9, namely HNH and RuvC domain (D10A/H840A), making it unable to undergo splicing and only having DNA binding function.[103] Huang et al.[42] developed a single-

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base editing system in *A. niger* based on this approach. The researchers used the *tef* promoter to drive the expression of deaminase (rat APOBEC1), nCas9 and UGI fusion proteins, in which rAPOBEC1 deaminase was linked to the N -end of nCas9 protein via a 16-aa XTEN linker and UGI was linked to the C -end of nCas9 protein via an SGGS linker. Based on these operations, a CBE expression cassette was constructed. This study successfully achieved single-base editing in *A. niger* with editing efficiencies between 47% and 100%, which is more efficient than that in animals and plants and will greatly facilitate precise molecular breeding in filamentous fungi. Furthermore, Zhou et al.[104] combined the tRNA-gRNA array expression system with CBE to form a multi site base editor (MBE), which inactivated core genes in eight natural product gene cluster in *A. nidulans*, with the inactivation efficiency of 40%.

3.3. Multiple Editing. The core problem of multiple knockouts lies in multiple targeting sites; therefore, the problem of how to construct multiple gRNA expression cassettes needs to be addressed. Among them, the most classic approach is that Liu et al.[48] introduced vectors containing Cas9 expression cassettes into recipient fungal cells through AMT, and then multiple mature gRNAs synthesized *in vitro* were introduced into Cas9-positive cells via PMT to achieve multiple gene editing. This method has been applied to *A. fumigatus*,[43] *A. niger*,[105] and *B. bassiana*.[106]

Besides, the tRNA-spacer system can also perform multi-target editing (Figure 5A), which is based on the principle that RNase P and RNase Z can release mature tRNAs from larger pre-tRNA transcripts due to the tRNA splicing mechanism.[107] In the CRISPR system, it is possible to use this mechanism to construct multiple sgRNAs so that multiple edits can be performed. Nødvig et al.[100] then expressed tandem sgRNAs separated by tRNA entities through the U3 promoter, thereby achieving multiplexing. Specifically, different gRNAs are linked by the 5′ and 3′ splice sites of tRNAs, with a tRNA spaced between the gRNAs, and then, the mature gRNAs are released by the action of RNase P and RNase Z.

Kujoth et al.[108] used two strategies to express dual sgRNAs. The first strategy is that two different sgRNA cassettes are linked in tandem by a linker sequence and expressed under the control of the same promoter and terminator. The other is that the sgRNA cassettes are expressed under the control of separate promoters and terminators, respectively (Figure 5C). The results show that separate promoters drive different sgRNAs with a better editing efficiency (73%).

Miao et al.[93] provided a CRISPR-based multigene recyclable editing system that works in a variety of filamentous fungi. They first constructed a strain of *ura3* gene-deficient *Trichoderma guizhouense* and transformed it with a donor fragment containing the *ura3* gene, enabling unrestricted gene cycle editing by two independent homologous recombination events for the specific integration of the knockout fragment and the traceless removal of the marker gene. Next, they knocked out *Ku70* and *Ku80* genes to eliminate the effect of NHEJ, and introduced the CRISPR system to target site-specific cleavage, which greatly increased the frequency of homologous recombination (90 times). Further, the authors performed four-point editing and found that all 4 sgRNAs worked when edited individually, but when simultaneous editing was performed, it was found that 4 sgRNAs would interfere with each other and lead to the inability of editing at some particular sites. For this purpose, the authors designed a method to insert selected markers only at the weakest editing efficiency points. Compared with the traditional method, only one selected marker was used for editing 4 positions, and this selected marker can be eliminated by a simple backscreening to achieve the purpose of trace-free editing. This study provides an effective one-step traceless editing strategy for gene function studies as well as a synergistic editing strategy for recoverable markers of multiple genes in filamentous fungi.

In addition to cas9 nucleases, studies on gene editing in filamentous fungi using Cas12a (Cpf1) nucleases, which are also type II, have increased in recent years.[109] Unlike Cas9, Cas12a (Cpf1) nuclease relies on a single crRNA guide for DNA localization, and the PAM in the recognized target gene is located at the 5′ end of the protospacer sequence, typically TTTN. Cas12a (Cpf1) cleaves at the distal PAM end of the protospacer sequence and produces sticky ends, which is the opposite of Cas9, which cleaves at the proximal PAM end to produce blunt ends. Due to these properties, Cas12a (Cpf1) can recognize multiple tandem crRNAs in a simple transcript, thus enabling the operation of simultaneous editing of multiple genes. Previously, Liu et al.[33] established the CRISPR/Cas9 system in *M. thermophila*, but there are still some limitations, including the need to design independent expression cassettes and a limited number of selected markers when targeting multiple genes. Therefore, in subsequent studies,[38] they established a CRISPR/Cas12a-based multigene editing tool in *M. thermophila* to address to the above-mentioned problems. First, to verify whether CRISPR/Cas12a works in the strain, the researchers performed a single knockout targeting the *amds* gene, as well as the *cre1* gene, and found that the knockout efficiency was over 90%. Subsequently, the researchers performed multiple gene knockout experiments by simultaneously targeting three genes, *cre-1*, *res-1*, and *gh1-1*, in which crRNAs were sequentially arranged for independent expression. It was found that the triple gene editing efficiency of the CRISPR/Cas12a system was similar to that of the CRISPR/Cas9 system, both around 30−40%, but the crRNA sequential arrangement was more convenient in constructing plasmids. In addition, to address the problem of a limited number of selected markers, the researchers developed a simple method for selected marker recovery. In simple terms, after integrating selected marker *neo* in the first round of screening, another new selected marker *bar* is introduced and *neo* is knocked out in the second round, then *neo* is introduced again and *bar* is knocked out in the third round, therefore recycling the *neo* and *bar* selected markers. With this approach, the researchers edited nine genes in the cellulase-producing strain after three rounds of experiments to obtain a strain M9 with a 9.0-fold and 18.5-fold increase in cellulase yield and lignocellulose activity. This approach can be used not only in CRISPR/Cas12a, but also in the previously established CRISPR/Cas9 with similar editing efficiencies.

4. CONCLUSION AND PERSPECTIVES

Although the CRISPR/Cas9 system has proven to be a powerful gene editing system in filamentous and macrofungi, there are still many challenges and deficiencies. Most importantly, since macrofungi and some filamentous fungi are heterokaryotic, if a transformant containing a resistance selected marker is not integrated into both nuclei, subsequent transformants will no longer contain the resistance marker due to division dilution,[117] which will greatly increase the difficulty of screening. With the help of AMA1, an autonomous

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replication element, *Aspergillus* spp. can achieve deletion of homozygous mutations of entire gene clusters,[2] but, since such autonomous replication elements have not been identified in the majority of macrofungi, more studies have had to retreat to the use of monokaryote of target species as starting strains to establish CRISPR systems. Nevertheless, monokaryons tend to differ significantly from heterokaryons in terms of growth morphology and type and content of metabolites,[118] which is clearly an obstacle when applying them in production. It has also been reported that monokaryons integrate CRISPR components less frequently than heterokaryons.[119] However, it is also a good idea to improve the transformation efficiency of multinucleated spores by controlling cell division and the mitotic cycle with chemical reagents. Additionally the targeting efficiency of gRNA affects the editing efficiency of CRISPR. Considering the various complex intracellular physiological and biochemical environments faced by gRNAs *in vivo*, such as the secondary structures of different gRNAs, substrate affinity to target sites, half-life, etc., all of them have a certain impact on the editing efficiency. Therefore, the design of multiple gRNAs for simultaneous editing at a target gene is also a solution. Moreover, constitutively expressed Cas9 is toxic, presumably related to accidental chromosome cleavage or off-target effect by overexpressed Cas9,[59,61,120] while selective use of ribozymes, use of weak promoters or repressive riboswitches to control Cas9 transcription or translation, and post-modification of Cas9 can ameliorate Cas9 toxicity. To minimize Cas9 toxicity and off-target effects, transient expression of the Cas9 protein is desirable, which can be achieved by controlling Cas9 expression with an inducible promoter or by transient plasmid proliferation. Since small phage-encoded anti-CRISPR proteins (Acrs) can inhibit the biological function of Cas9,[121] an efficient CRISPR-Cas9 genome editing tool has been developed in *Streptomyces* by introducing the Cas9 inhibitor AcrIIA4, which significantly improved Cas9 editing efficiency.[122] This Cas9 inhibitor model can also be applied to gene editing operations in filamentous and macrofungi. Therefore, the application of the CRISPR/Cas9 system in filamentous and macrofungi is still immature, and the editing efficiency is not high enough in most nonmodel species. Further elucidation of the metabolic regulatory mechanisms of nonmodel fungi and the internal mechanism of action of the CRISPR/Cas9 system is still to be elucidated, which will help to further expand its application in fungi.

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J.-Y. Shen prepared the figures/tables and wrote the manuscript. J.-Y. Shen, Q. Zhao, and Q.-L. He revised the manuscript.

Notes

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